



Course	: Diploma in Electronic Systems (EGDF18) Diploma in Electronic & Computer Engineering (EGDF20)
Module	: Embedded System Design & Technology (EGE320)
Tutorial No	: Tutorial 5
Tutorial Title	: Touchscreen Panel
Objective	: Describe the touch system hardware components by recognizing the properties and purposes of each component in the system.

Questions:

1. How does a touchscreen panel being layered together with a LCD screen?
 - A. It is layered on top of a LCD screen.
 - B. It is layered at the bottom of a LCD screen.
 - C. It is layered in between a LCD screen.
 - D. It is layered both top and bottom of a LCD screen.
2. Which of the following is **NOT** a touchscreen panel sensing technology?
 - A. Resistive
 - B. Capacitive
 - C. Inductive
 - D. Infrared
3. What is the use of spacer dot found in between the top and bottom transparent sheets coated with ITO in a resistive technology sensing structure?
 - A. To prevent the top and bottom sheets from contacting when pressed.
 - B. To prevent the top and bottom sheets from contacting when not pressed.
 - C. To perform resistive sensing by measuring voltages.
 - D. To fill the gap between the top and bottom sheets.

4. How would the touch point voltage be measured when voltage is imposed between the electrode X1 (0 v) and X2 (5 v) on the top sheet?
 - A. Through electrode X1
 - B. Through electrode X2
 - C. Through Spacer Dot
 - D. Through electrode Y
5. What is the possible voltage measured if the touch point is nearer to electrode X2? Assuming voltage is imposed between the electrode X1 (0 v) and X2 (5 v) on the top sheet.
 - A. 1 v
 - B. 2 v
 - C. 3 v
 - D. 4 v
6. Which of the following environments is projected capacitive sensing best suited for?
 - A. Wet or underwater
 - B. High-vibration industrial applications
 - C. Dusty environments
 - D. Clean, dry environments with bare-finger touch
7. If the center of a resistive touchscreen panel is touched, what would be the voltage measured at both X and Y electrodes given the maximum voltage imposed to the electrodes is 5 v?
 - A. 1.5 v
 - B. 2.5 v
 - C. 3.5 v
 - D. 4.5 v
8. How does the X and Y electrodes being arranged in the Projected Capacitive Structure?
 - A. Parallel
 - B. Series
 - C. Intersecting in matrix
 - D. Series for X electrode and parallel for Y electrode
9. How does a human finger affect the electrostatic capacitance between the X and Y electrodes in the Projected Capacitive Sensing?
 - A. Human finger introduces additional capacitance.
 - B. Human finger introduces additional resistance.
 - C. Human finger introduces additional inductance.
 - D. Human finger introduces additional pressure.
10. What is the role of the touch controller which connects between the touch sensor and microcontroller?

- A. Captures the voltage or signal change on the touchscreen panel.
- B. Allows the touchscreen software driver code running inside to know how to interpret the touch event information.
- C. Takes information from the touch sensor and translate it into information that microcontroller can understand.
- D. Fetches data from the frame buffer, decodes it to the required bit format and feeds it to the microcontroller.